Group 1 Assignment

Modems & RS-232 Interfacing Sequences and Line Coding & Its Techniques

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Course: CIT 3212 — Computer Networks
Topic Area: The Physical Layer (OSI Layer 1)
Assignment Reference: “Assignment on Physical Layer (Group 1)”

Prepared as a written answer / presentation guide

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1. Introduction

Both parts of this assignment belong to the **Physical Layer (Layer 1)** of the OSI model — the layer that moves raw bits across a transmission medium.

- **Modems and RS-232** are about how data is *carried between two devices*: the modem converts digital data into signals suitable for a transmission line, while RS-232 defines the electrical and procedural interface that lets a computer talk to a modem (or another serial device).
- **Line coding** is about how a sequence of bits is *converted into a digital signal* that can be placed on the medium.

2. Modems

2.1 Definition and purpose

A **modem** (short for **MOdulator–DEModulator**) is a device that converts digital data from a computer into an analogue signal that can be transmitted over a communication line, and converts received analogue signals back into digital data.

Computers produce digital (square-wave) signals, but many transmission media — ordinary telephone lines, cable TV coax, radio links — carry **analogue** signals. A modem bridges that gap and is the classic example of a Layer-1 device.

- **Modulation** — the sender changes (modulates) a carrier

wave to represent the digital 0s and 1s.

- **Demodulation** — the receiver extracts (demodulates) the original bits from the incoming carrier.
- Most modems are **full-duplex**: they modulate and demodulate at the same time in both directions.

2.2 How a modem works

1. Digital data (bits) arrives from the computer over the serial interface (e.g. RS-232).
2. The modem's **modulator** impresses the bits onto a carrier signal using a modulation technique (ASK, FSK, PSK or QAM).
3. The modulated analogue signal travels over the transmission line (e.g. telephone line).

4. At the far end, the **demodulator** recovers the bits from the carrier.
5. The bits are passed to the receiving computer through its serial interface.

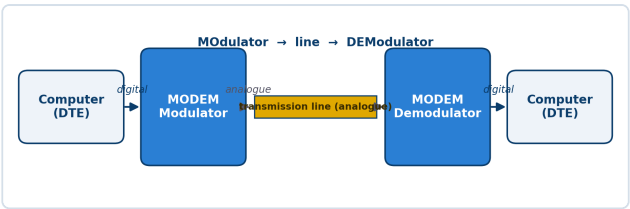


Figure 1 — A modem modulates digital data onto the analogue line and demodulates it at the far end.

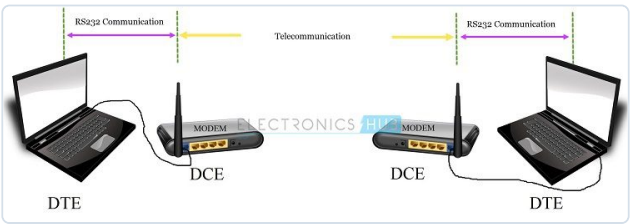


Figure 2 — A laptop connected to a modem through an RS-232 serial cable (real-world setup).

> A **carrier** is a high-frequency sine wave with three properties that can be varied

to carry data: **amplitude**, **frequency** and **phase**.

2.3 Types of modems

By direction

Type: Half-duplex
Description: Transmits in both directions but not at the same time.

By direction

Type: Full-duplex
Description: Transmits in both directions simultaneously.

By transmission medium

Type: Dial-up (PSTN) modem
Description: Uses ordinary telephone lines; up to 56 kbps.

By transmission medium

Type: DSL / ADSL modem
Description: High-speed data over telephone lines without blocking voice.

By transmission medium

Type: Cable modem
Description: Data over cable-TV (coaxial) networks.

By connection

Type: Internal modem
Description: Expansion card inside the computer.

By connection

Type: External modem
Description: Separate box connected by a serial cable (e.g. RS-232).

By carrier type

Type: Wireless / cellular modem
Description: Uses radio (GSM/4G/5G) as the transmission medium.

2.4 Modulation techniques used by modems

ASK — Amplitude Shift Keying

Property varied: Amplitude
Description: Carrier amplitude is switched between two levels to represent 0 and 1.

FSK — Frequency Shift Keying

Property varied: Frequency
Description: Two (or more) frequencies represent the bits; used in early low-speed modems.

PSK — Phase Shift Keying

Property varied: Phase
Description: The phase of the carrier is shifted to encode bits; more robust than ASK.

QAM — Quadrature Amplitude Modulation

Property varied: Amplitude + Phase

Description: Combines ASK and PSK to send many bits per symbol; used by high-speed modems.

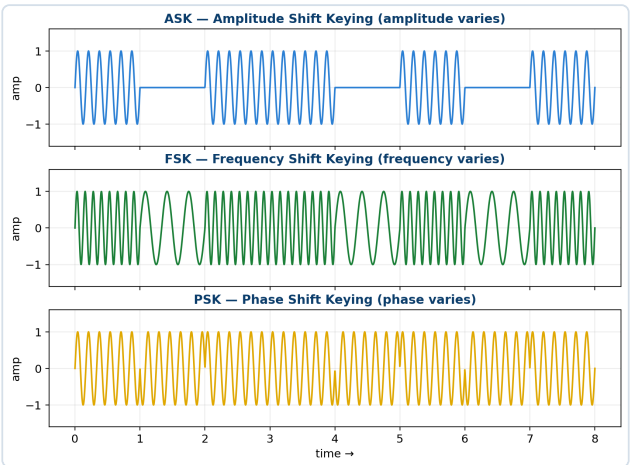


Figure 3 — The same bits encoded by changing amplitude (ASK), frequency (FSK) or phase (PSK).

Because a symbol can carry several bits, modem speed is measured in **bits per second (bps)**: `bit rate = baud rate × bits per symbol`.

2.5 Standards and data-rate evolution

V.22 / V.22bis

Speed: 1200 / 2400 bps
Notes: Early dial-up standards.

V.32 / V.32bis

Speed: 9.6 / 14.4 kbps
Notes: Introduced echo cancellation and trellis coding.

V.34

Speed: 28.8 – 33.6 kbps
Notes: Adaptive modulation.

V.90 / V.92

Speed: 56 kbps downstream
Notes: Digital downstream, analogue upstream.

ADSL / VDSL

Speed: Mbps range
Notes: Broadband over telephone lines.

3. RS-232 Interfacing Sequences

3.1 What is RS-232?

RS-232 (Recommended Standard 232, also EIA-232) is a widely used standard for **serial (one bit at a time) binary data interchange** between Data Terminal Equipment (DTE) and Data Communication Equipment (DCE) using a single-ended, point-to-point connection.

It defines three things: the **electrical** characteristics (voltage levels), the **mechanical** interface (connectors and pin assignments) and the **procedural** sequence of signals

used to set up, transfer and end a communication.

- **DTE** — Data Terminal Equipment, e.g. a computer or terminal (source/sink of data).
- **DCE** — Data Communication Equipment, e.g. a modem (the device that talks to the network).

3.2 Electrical characteristics and voltage levels

RS-232 uses **negative logic** and large voltage swings to resist noise:

Logic state	Signal name	Voltage at transmitter
0 (Space)	ON / active	+3 V to +15 V (typically +12 V)
1 (Mark)	OFF / idle	−3 V to −15 V (typically −12 V)

- Voltages between −3 V and +3 V are undefined and rejected as noise.

- Maximum cable length ≈ **15 m (50 ft)** at 19.2 kbps; longer at lower speeds.
- Common data rates up to **20 kbps** (115.2 kbps over short cables).
- It is **single-ended** (unbalanced) — each signal is referenced to a common ground, limiting noise immunity and distance versus RS-422/RS-485 (differential).

3.3 Connectors and pin functions

RS-232 uses **DB-25** (25-pin) or the more common **DB-9** (9-pin) connector.

1

DB-25 pin: 8
Name: DCD — Data Carrier Detect
Direction: DCE → DTE
Function: Modem tells the computer it detected a carrier from the far end.

2

DB-25 pin: 3
Name: RXD — Receive Data
Direction: DCE → DTE
Function: Data received by the DCE, sent to the DTE.

3

DB-25 pin: 2
Name: TXD — Transmit Data
Direction: DTE → DCE
Function: Data transmitted from the DTE to the DCE.

4

DB-25 pin: 20
Name: DTR — Data Terminal Ready
Direction: DTE → DCE
Function: Computer says it is powered and ready.

5

DB-25 pin: 7
Name: GND — Signal Ground
Direction: —
Function: Common reference for all signals.

6

DB-25 pin: 6
Name: DSR — Data Set Ready
Direction: DCE → DTE
Function: Modem says it is powered and ready.

7

DB-25 pin: 4
Name: RTS — Request To Send
Direction: DTE → DCE
Function: Computer asks to send data.

8

DB-25 pin: 5
Name: CTS — Clear To Send
Direction: DCE → DTE
Function: Modem grants permission to send.

9

DB-25 pin: 22
Name: RI — Ring Indicator
Direction: DCE → DTE
Function: Modem signals an incoming call (ring).

Three signal “pairs” to remember: **TXD/RXD** (data), **DTR/DSR** (device readiness) and **RTS/CTS** (flow/handshake).

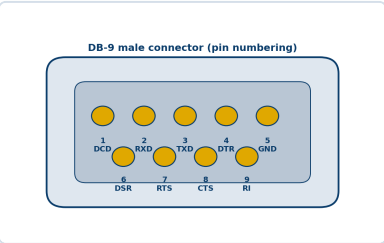


Figure 4 — DB-9 connector pin numbering (5 pins on top, 4 below).

3.4 The RS-232 interfacing (handshaking) sequence

Before data can flow, the two ends exchange control signals in a defined order — the “interfacing sequence”. A typical dial-up connection:

```
STEP 1  DTE asserts DTR   ->
Computer: "I am on and ready."
STEP 2  DCE asserts DSR   ->
Modem:    "I am on and ready."
STEP 3  DTE asserts RTS   ->
Computer: "I want to send data."
STEP 4  DCE asserts CTS   ->
Modem:    "Go ahead, send."  [Data
flows on TXD]
        (for incoming data the modem
raises DCD; data arrives on RXD)
STEP 5  Data exchange     ->  TXD
(send) / RXD (receive)
STEP 6  DTE drops RTS     ->
Computer: "I have finished sending."
STEP 7  DCE drops CTS     ->
Modem:    "Acknowledged."
STEP 8  DTE/DCE drop DTR/DSR ->
Connection released.
```

Ring Indicator (RI): when a call arrives, the modem pulses RI. The computer then raises DTR, the modem answers and raises DCD, and data transfer begins.

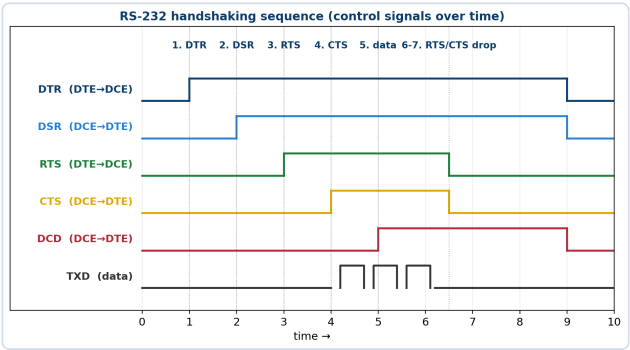


Figure 5 — Timing view of the handshake: DTR/DSR, then RTS/CTS, then data on TXD.

3.5 Flow control and the null modem

- **Hardware flow control (RTS/CTS):** the receiver lowers CTS to tell the sender to pause.
- **Software flow control (XON/XOFF):** special characters (Ctrl-Q / Ctrl-S) are embedded in the data stream instead of using control wires.
- **Null modem:** a cable/adapter connecting two *DTE* devices directly. It **crosses** TXD↔RXD and RTS↔CTS (and often loops

DTR↔DSR) so two DTEs can communicate without a modem.

3.6 Limitations of RS-232

- Short distance (≈15 m) and low speed.
- Single-ended signalling — poor noise immunity.
- Point-to-point only (one device per port).
- Large voltage swings and many wires make it bulky.
- Largely replaced by **USB** and differential standards such as RS-485, though still used for industrial/embedded devices.

4. Line Coding and Its Techniques

4.1 Definition and purpose

Line coding is the process of converting a sequence of bits (digital data) into a **digital signal** that can be transmitted over a baseband channel. It maps binary 1s and 0s to voltage or current levels on the line.

Line coding happens *before* modulation: it turns bits into a digital waveform. If the medium is analogue, that digital signal is then further modulated (*carrier modulation*).

4.2 Why line coding matters — design goals

Goal	Meaning
No DC component / no baseline wandering	Avoid a non-zero average voltage that drifts and confuses the receiver. Good codes have no DC component.
Self-synchronisation	The signal must contain enough transitions for the receiver to recover the clock and stay in step.
Error detection	Some codes build in the ability to detect errors.
Noise immunity	Resistance to interference on the line.
Bandwidth efficiency	Use as little bandwidth as possible for a given data rate.

4.3 Line coding schemes

Schemes are grouped into five broad families: **Unipolar, Polar, Bipolar, Block coding** and **Scrambling**.

(a) Unipolar

All signal levels are on one side of the time axis (either positive or negative). The simplest example is **Unipolar NRZ**: a 1 is a positive voltage and a 0 is zero volts (or vice versa).

- ✔ Simple to implement.
- ✘ Has a DC component and poor synchronisation (long runs of the same bit cause loss of clock).

(b) Polar

Uses both positive and negative voltages. Main variants:

NRZ-L (Non-Return-to-Zero Level)

How a 1 / 0 is represented: Level depends on bit: +V for 1, −V for 0.
Notes: Simple; but long runs of the same bit cause baseline wandering.

NRZ-I (NRZ Invert)

How a 1 / 0 is represented: A 1 causes a transition (inversion) at the start of the bit; a 0 causes no change.
Notes: Better synchronisation for long runs of 1s; a 0-run is still a problem.

RZ (Return-to-Zero)

How a 1 / 0 is represented: Signal returns to zero in the middle of each bit.
Notes: Self-synchronising, but needs 2× bandwidth.

Manchester

How a 1 / 0 is represented: A transition in the middle of each bit: high→low for 0, low→high for 1 (or the reverse).
Notes: Fully self-synchronising; used by classic Ethernet (10Base-T).

Differential Manchester

How a 1 / 0 is represented: A transition in the middle of every bit; a 0 also has a transition at the start of the bit.
Notes: Self-synchronising; used by Token Ring.

(c) Bipolar

Uses three voltage levels: positive, zero and negative. A 0 is always zero volts; 1s alternate in polarity (this is **AMI — Alternate Mark Inversion**).

- No DC component and good bandwidth efficiency.

- In **Pseudoternary**, the roles are swapped: 1s are zero volts and 0s alternate in polarity.

(d) Block coding

Instead of coding bit-by-bit, a block of m bits is replaced by a larger block of n bits ($n > m$) to guarantee transitions and remove DC. Common schemes include **4B/5B** (4 data bits → 5 code bits) and **8B/10B**. Block coding is often combined with NRZ-I (e.g. 100Base-FX Fast Ethernet uses 4B/5B + NRZ-I).

(e) Scrambling

Used for long-distance transmission where block coding would add too much overhead. Scrambling modifies long runs of the same bit to create transitions without adding bits. Common techniques are **B8ZS** (Bipolar with 8-Zero Substitution) and

HDB3 (High-Density Bipolar 3-zero).

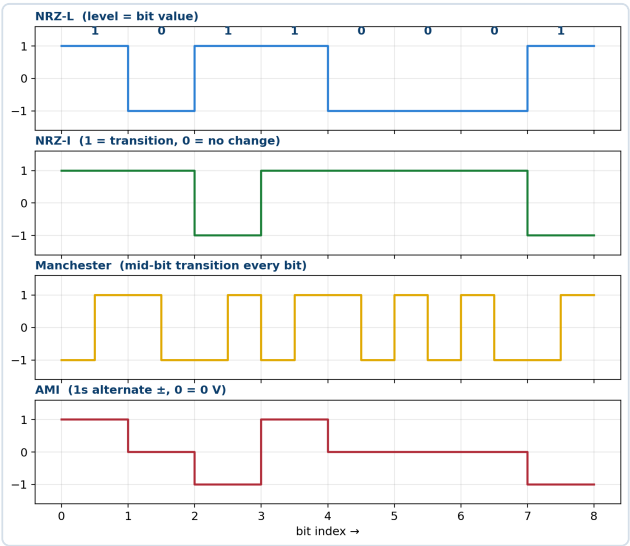


Figure 6 — The same data bits (1 0 1 1 0 0 0 1) encoded with NRZ-L, NRZ-I, Manchester and AMI.

4.4 Worked example

Data bits: `1 0 1 1 0 0 0 1`

```
NRZ-L :  +  -  +  +  -  -  -  +
NRZ-I :  ^  .  ^  .  .  .  .  ^
(^ = transition at start of bit; .
= none)
Manchester: each bit has a mid-bit
transition (e.g. 1 = low->high, 0 =
high->low)
AMI   :  +  0  -  +  0  0  0  -
(1s alternate +/-, 0 = 0 V)
```

When drawing these for a presentation, remember: NRZ-L follows the level; NRZ-I follows transitions; Manchester always has a mid-bit transition; AMI alternates polarity on 1s.

4.5 Comparison of line coding schemes

Unipolar NRZ

Levels: 2 (0, +V)
DC component: Yes
Self-sync: No
Bandwidth: Low
Typical use: Simple/legacy

NRZ-L

Levels: 2 (−V, +V)
DC component: Possible
Self-sync: No
Bandwidth: Low
Typical use: General digital

NRZ-I

Levels: 2 (−V, +V)
DC component: Possible
Self-sync: Partly
Bandwidth: Low
Typical use: USB, Fast Ethernet (with 4B/5B)

RZ

Levels: 3
DC component: No
Self-sync: Yes
Bandwidth: High (2×)
Typical use: Legacy

Manchester

Levels: 2
DC component: No
Self-sync: Yes
Bandwidth: High
Typical use: 10Base-T Ethernet

Diff. Manchester

Levels: 2
DC component: No
Self-sync: Yes
Bandwidth: High
Typical use: Token Ring

AMI

Levels: 3
DC component: No
Self-sync: Partly
Bandwidth: Moderate
Typical use: Long-distance T1 lines

4B/5B + NRZ-I

Levels: 2
DC component: No
Self-sync: Yes
Bandwidth: Moderate
Typical use: 100Base-FX

5. Summary

- A **modem** converts digital data to analogue (modulation) and back (demodulation), using ASK, FSK, PSK or QAM

over media such as telephone lines.

- **RS-232** is a standard serial interface defining voltage levels, connector pin-outs and a handshaking sequence (DTR/DSR, RTS/CTS, DCD, RI) between a DTE and a DCE.
- **Line coding** converts bits into a digital signal. Good codes have no DC component, are self-synchronising, and resist noise. Families include unipolar, polar (NRZ, RZ, Manchester), bipolar (AMI), block coding (4B/5B) and scrambling (B8ZS, HDB3).

Suggested presentation slides

1. Title & scope (Physical Layer)
2. What is a modem? Modulation vs demodulation
3. Modem types & modulation techniques (ASK/FSK/PSK/QAM)

4. RS-232: definition, DTE/DCE, voltage levels
5. RS-232 pin-out (DB-9) and signal pairs
6. RS-232 handshaking sequence (step diagram)
7. Null modem & limitations of RS-232
8. Line coding: purpose and design goals
9. Line coding schemes (unipolar → polar → bipolar)
10. Block coding & scrambling
11. Worked waveform example & comparison table
12. Summary & references

Prepared in answer to the Group 1 assignment under “Assignment on Physical Layer (Group 1)” — CIT 3212 Computer Networks.